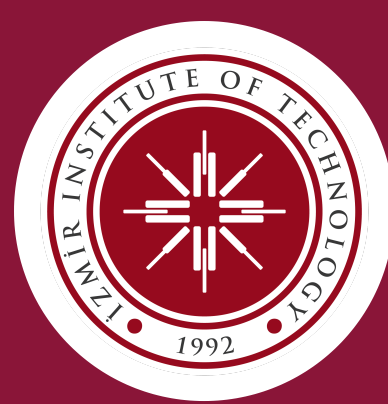




A Systematic Comparison of Large and Small Language Models in Standalone, Hybrid & Multi-Agent Architectures

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1 Abstract / Summary

Measurement-first platform benchmarking **ten architectures in three families** across **five public benchmarks** under a unified observability contract (accuracy · latency · cost · energy · carbon). We introduce **EATS** (Efficiency-Accuracy Trade-off Score) as an architecture-agnostic metric for comparing quality and efficiency together. Routing reaches **EATS 0.893** on HellaSwag and **0.854** on TruthfulQA; Pure Swarm leads GSM8K at **EATS 0.841**; Speculative achieves **EATS 0.908** on ARC. Novel **Blackboard-style** and **swarm** designs show task-specific strengths on reasoning benchmarks.

2 Background / Literature Review

LLMs offer stronger reasoning but impose higher latency, cost, and energy. **SLMs** are cheaper and easier to self-host, yet usually trail on broad reasoning tasks [1][2].

Our review of **54 collaborative SLM–LLM studies** shows a persistent gap: **accuracy is usually reported, but cost and energy rarely are** [10].

RQ1 Performance. How do collaborative systems compare with standalone LLMs on reasoning and classification?

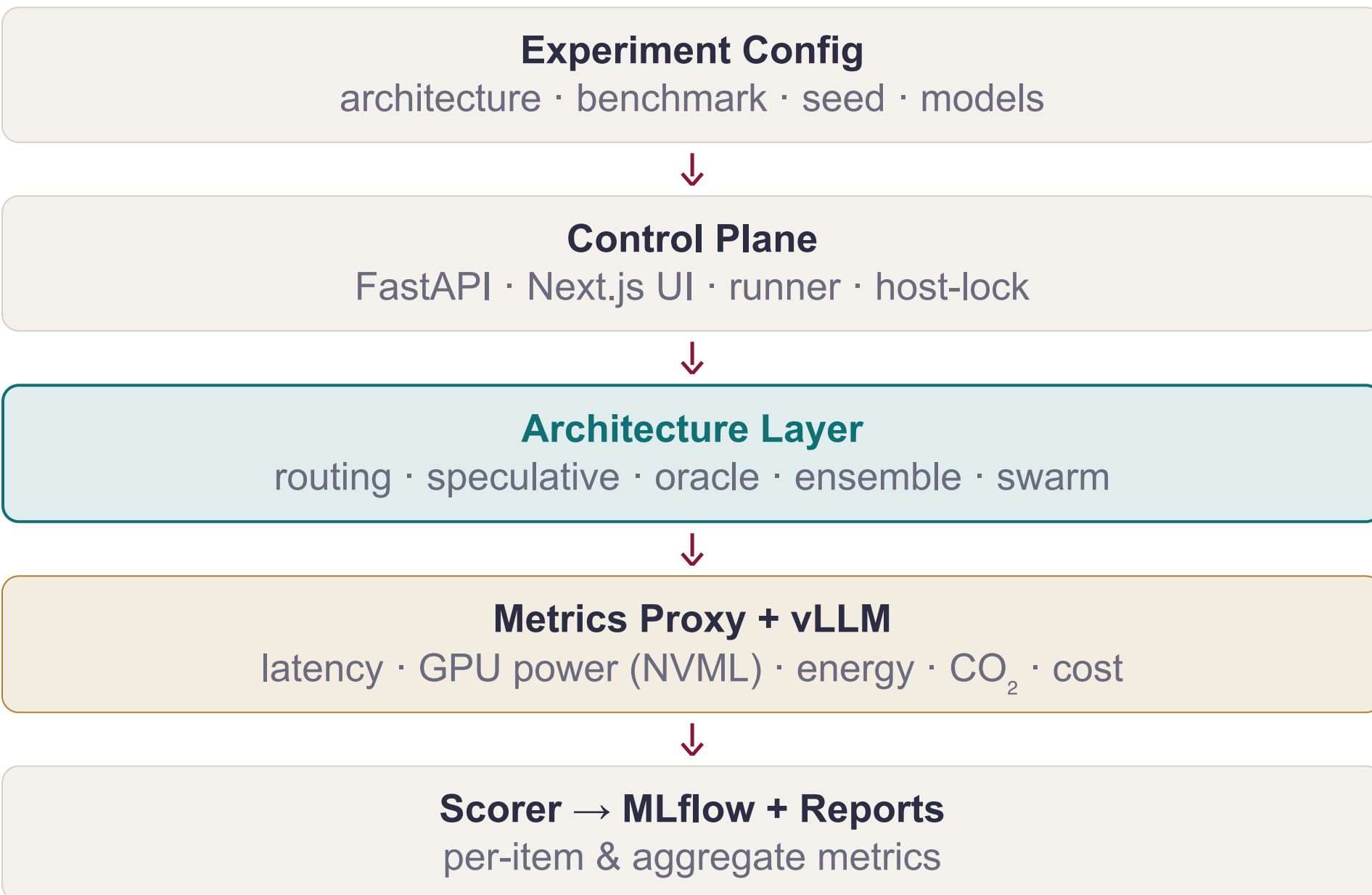
RQ2 Efficiency. How do hybrid and multi-agent designs change cost, latency, and energy?

RQ3 Unified metric. Does EATS provide a stable cross-architecture trade-off ranking?

RQ4 Empirical grounding. Do measured results match the trends from the 54-study SLR?

3 Measurement-First Platform

Every architecture consumes the same Query and returns the same Response carrying **accuracy, latency, tokens, cost, energy & CO₂** per query. The architecture layer is the *only* interchangeable component, so differences are attributable to orchestration logic rather than model API shape [4].



TIER	HOST	VRAM	MODELS
SLM	NVIDIA L4	24 GB	gemma4, qwen3.5, llama3.2 (3–4B)
Mid LLM	NVIDIA RTX PRO6000	96 GB	gpt-oss, gemma4, qwen3.5 (20–35B dense + MoE)
Heavy LLM	NVIDIA H200	141 GB	gpt-oss, llama3.3 (70–120B)

4 EATS · Efficiency–Accuracy Trade-off Score

A single score comparing any architecture against the standalone-LLM baseline.

$$\text{EATS} = \frac{\text{Accuracy}}{\text{Accuracy} + \lambda \cdot P_{\text{efficiency}} + \beta \cdot P_{\text{accuracy}}}$$
$$P_{\text{efficiency}} = 0.65 \cdot \text{Cost} + 0.20 \cdot \text{Latency} + 0.15 \cdot \text{Energy}$$
$$P_{\text{accuracy}} = 1 - \text{Accuracy}$$

$\beta = 0.60$ $\lambda = 0.40$

all terms normalised per standalone-LLM baseline · $\beta = 0.60$, $\lambda = 0.40$ calibrated via grid search on MMLU validation split

5 Energy, Carbon & Cost

Per-request measurements use GPU power sampling (NVIDIA NVML) at the inference host [4].

$$\text{Cost} = \text{GPU rate} \cdot \text{time}$$
$$E = P_{\text{GPU}} \cdot \Delta t$$
$$\text{Latency} = \Delta t \text{ per request}$$
$$\text{CO}_2 = k \cdot E$$

$k = 380 \text{ gCO}_2/\text{kWh}$ (EU avg) · E in joules + 3.6×10^6

6 Benchmarks & Protocol

Five benchmarks span the **capability dimensions most cited in our companion SLR**, letting us test whether efficiency gains persist as tasks shift from knowledge retrieval to chain-of-thought math and adversarial fact-checking [5][6][7][8][9].

BENCHMARK	WHAT IT MEASURES
MMLU	Broad knowledge — 57 academic disciplines
ARC-C	Scientific reasoning — multi-step physical-world science
GSM8K	Mathematical reasoning — grade-school math, free-form
HellaSwag	Commonsense inference — sentence completion
TruthfulQA	Adversarial truthfulness — calibration vs. persuasive falsehoods

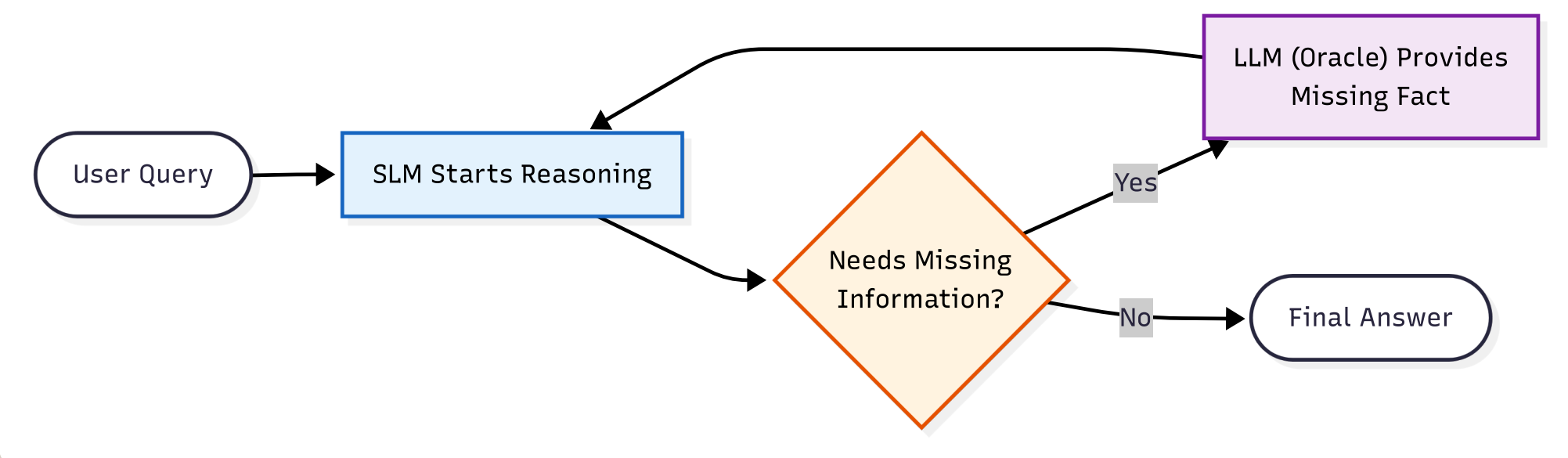
7 Architecture Taxonomy

Ten architectures in three families. † marks novel project contributions introduced in the current study.

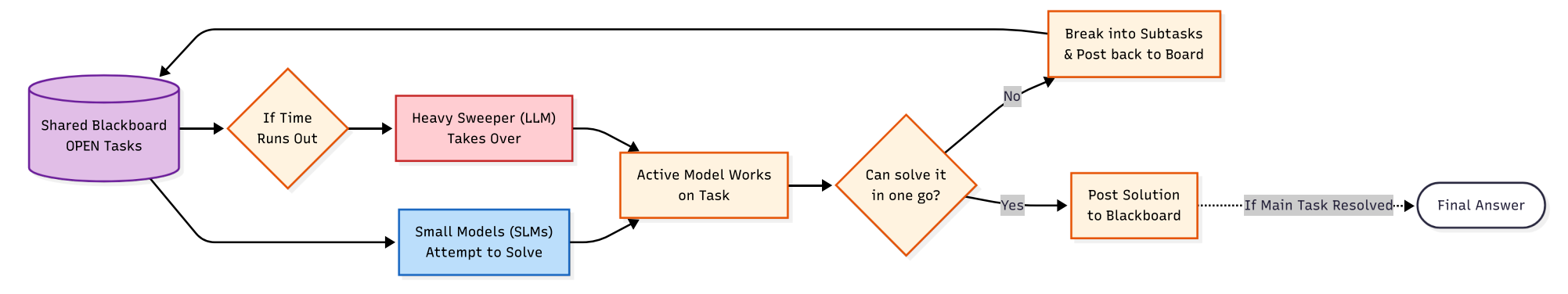
The taxonomy moves from **internal sparsity** and direct inference to **explicit coordination**, making it possible to ask whether efficiency comes from the model itself, from selective escalation, or from multi-step collaboration [1][2][3].

FAMILY	ARCHITECTURE	KEY MECHANISM
STAND	Single-Model	direct inference
STAND	MoE	internal sparse expert activation
HYBRID	Routing	confidence SLM→LLM escalation
HYBRID	Speculative	SLM drafts; LLM rewrites suffix
HYBRID	Debate (POA)	proponent–opponent–arbitrator
HYBRID	Active Oracle †	targeted LLM sub-queries
HYBRID	Blackboard †	bid-based task board; LLM sweeper fallback
HYBRID	Entropy Blackboard †	uncertainty-penalised bidding
MULTI	Ensemble	majority or confidence-weighted voting
MULTI	Pure Swarm †	SLM-only peer coordination; no LLM online

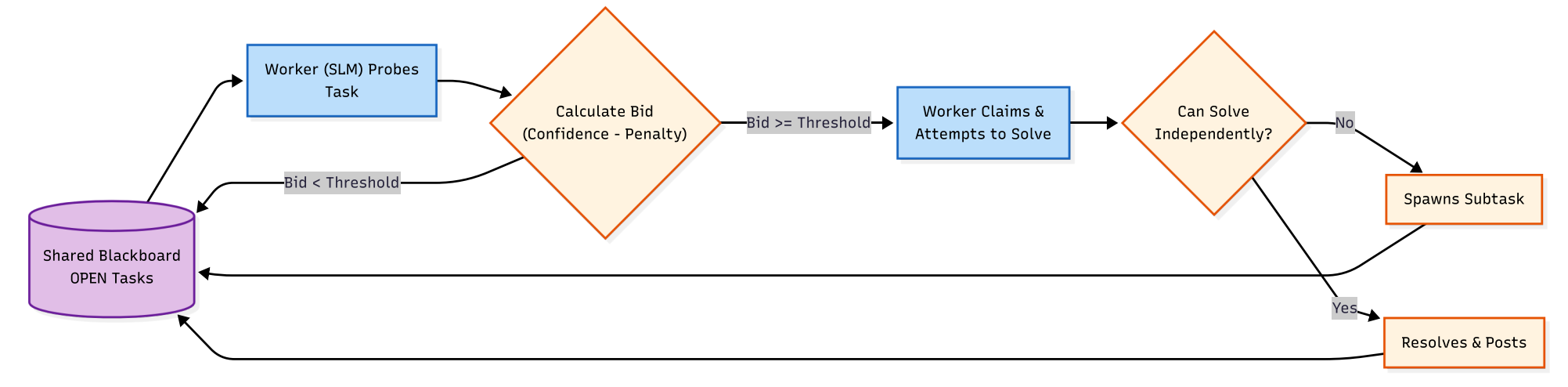
ACTIVE ORACLE



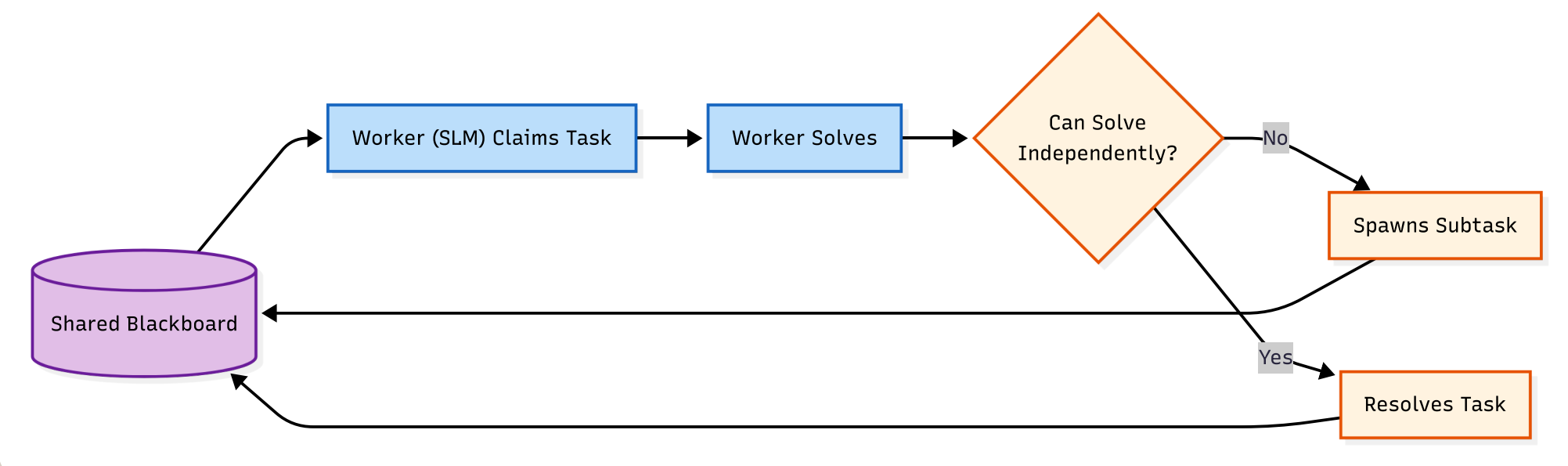
BLACKBOARD



ENTROPY BLACKBOARD



PURE SWARM



8 Model Selection & Architecture Pairing

We keep the comparison focused on **architecture behavior** by holding the core hybrid pair mostly fixed: a cheap **Gemma4-4B** worker and a stronger **Gemma4-31B** verifier/sweeper.

SELECTION RULE

Models were chosen for **family diversity, open-weight reproducibility, and MoE coverage**, so sparse-expert baselines can be compared against external orchestration.

BEST-WORKING PAIRINGS

Routing and **Speculative** with **Gemma4-4B → Gemma4-31B** gave the strongest EATS on **ARC, HellaSwag, and TruthfulQA**. The most efficient standalone baselines were the **Qwen3.5-35B-A3B** and **Gemma4-26B-A4B** MoE checkpoints.

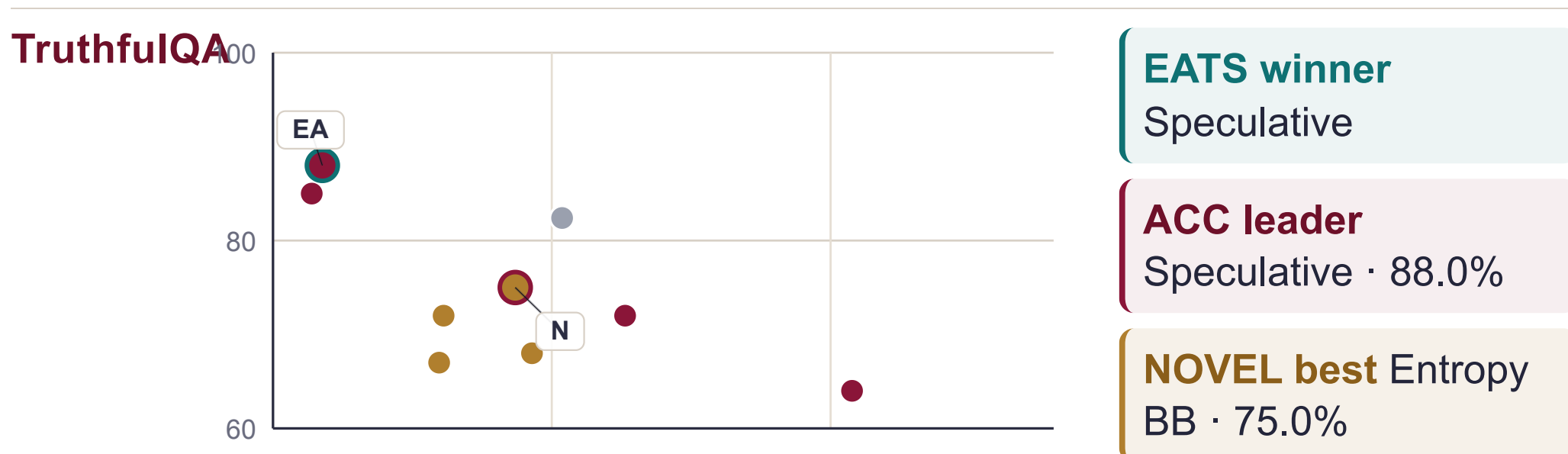
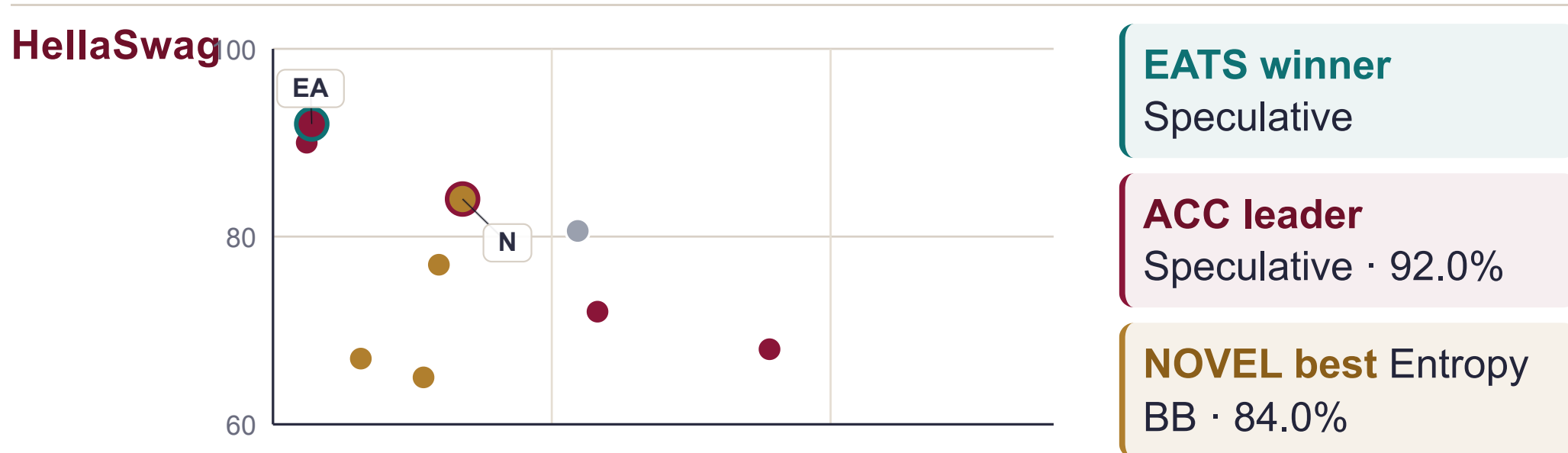
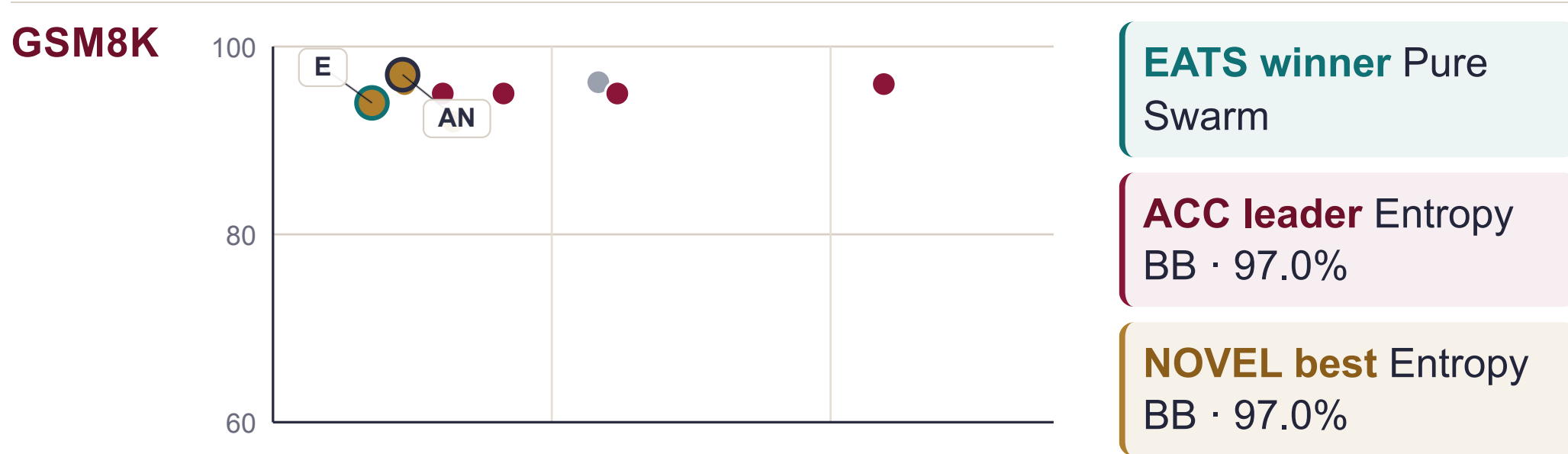
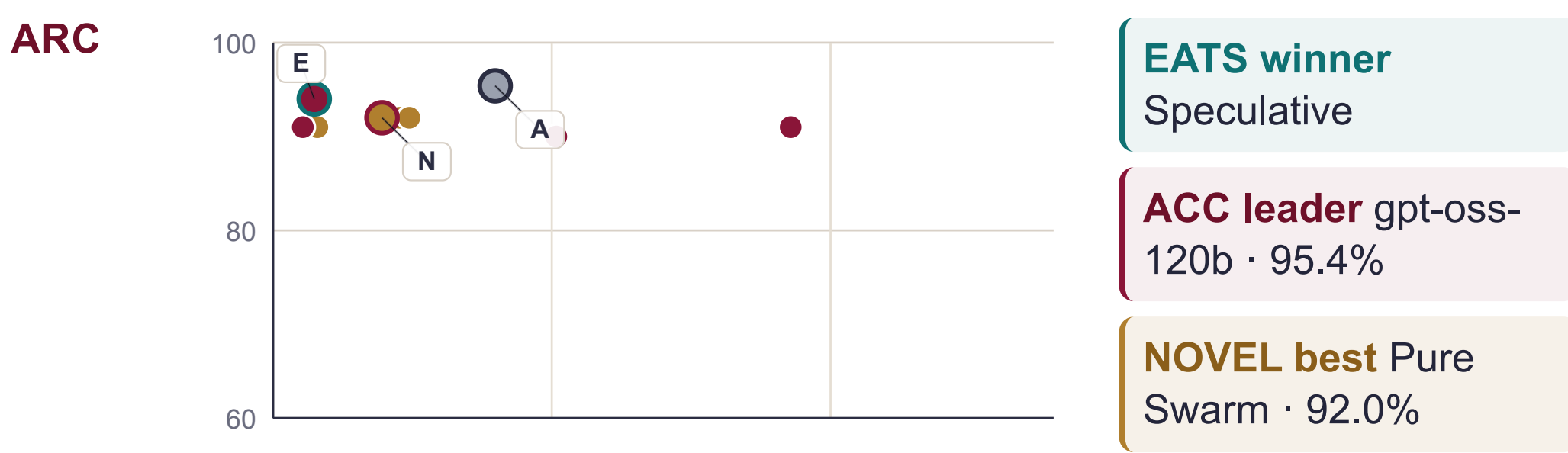
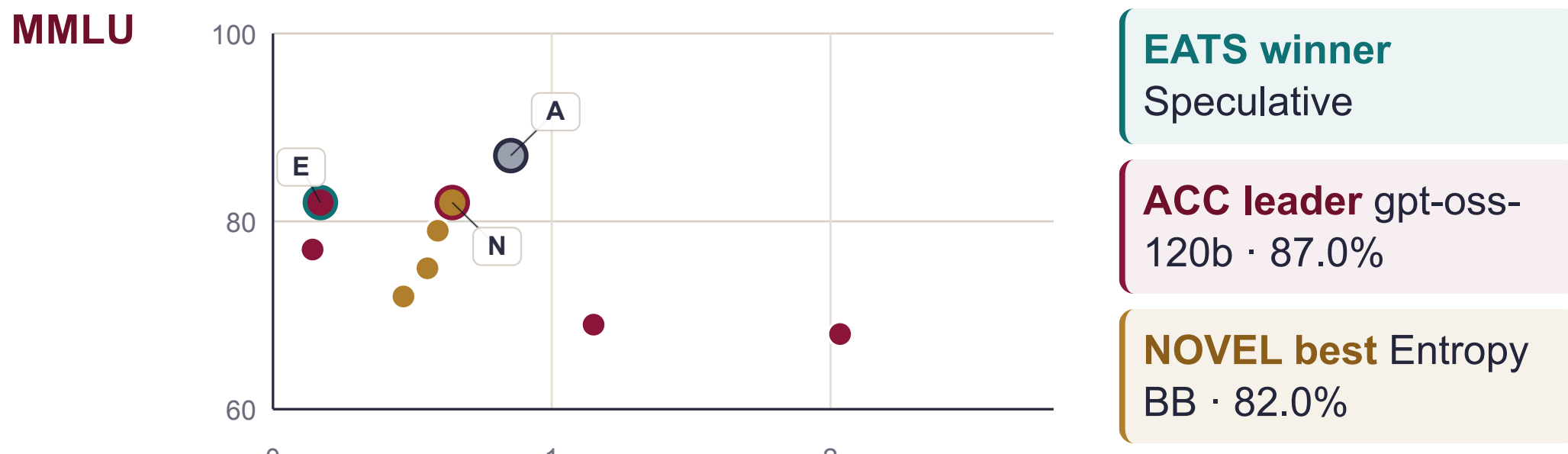
ARCHITECTURE	MODELS ONLINE	PURPOSE
Routing / Speculative	gemma4-4b → gemma4-31b	Cheap draft/router + stronger verifier for the best quality-efficiency frontier.
Oracle / Blackboard family	gemma4-4b workers + gemma4-31b sweeper	Controls for model family while changing only orchestration logic.
Debate / Ensemble / Pure Swarm	gemma4-4b, qwen3.5-4b, llama3.2-3b, ministral3-3b	Tests whether compact-model diversity and coordination can replace online LLM use.

Standalone baselines qwen3.5-35b-a3b, gemma4-26b-a4b, gpt-oss-120b, llama3.3-70b

Placing the same **Gemma4-4B → Gemma4-31B** pair inside routing, speculative, oracle, and blackboard-style designs is deliberate: the poster compares **control strategies** under a mostly fixed capability pair, while MoE baselines test whether external orchestration still adds value against sparse-expert standalone models [1][2] [4].

9 Benchmark Readouts

Each mini-plot shows accuracy against efficiency penalty, so the strongest trade-offs appear toward the upper-left. **E** = EATS winner, **A** = accuracy leader, **N** = best novel result.



10 EATS Landscape

Canonical EATS for the latest qualified run of each architecture; the standalone row shows the best MoE baseline on each benchmark.

ARCHITECTURE	MMLU	ARC	GSM8K	HSWAG	TQA
Routing	0.80	0.90	0.78	0.89	0.85
Speculative	0.82	0.91	0.72	0.90	0.86
Standalone (MoE)	0.68	0.73	0.66	0.59	0.61
Blackboard †	0.69	0.80	0.82	0.67	0.55
Entropy BB †	0.69	0.79	0.83	0.69	0.60
Active O. †	0.67	0.89	0.75	0.67	0.64
Debate POA	0.52	0.66	0.64	0.53	0.52
Pure Swarm †	0.67	0.82	0.84	0.60	0.61
Ensemble	0.40	0.53	0.52	0.43	0.38

Darker = higher EATS · bold = benchmark winner · tinted rows / † = novel contribution.

11 Conclusions

The main synthesis is not that one model wins everywhere, but that **the best architecture depends on how much capability you can afford to escalate online**. That is why the poster reports both raw accuracy and EATS rather than collapsing the study into a single leaderboard.

RQ1 Collaborative systems are competitive with standalone LLMs; the strongest results come from routing, speculative decoding, and novel coordination on reasoning-heavy tasks.

RQ2 Hybrid escalation gives the clearest cost and energy gains, while multi-agent coordination usually pays a latency penalty.

RQ3 EATS gives a consistent cross-architecture ranking: speculative leads four benchmarks, and Pure Swarm leads GSM8K.

RQ4 The measurements corroborate 8 of 10 SLR trends: SLM-first escalation is the strongest general pattern, while coordination and entropy-aware bidding help on task-specific cases.

LIMITATIONS

- Single hardware stack (L4 / PRO6000 / H200) — results may not generalise to other GPU tiers or cloud inference endpoints.
- Benchmark scope covers 5 NLP task types; code generation, retrieval-augmented and long-context tasks are not included.
- EATS weights ($\beta = 0.60$, $\lambda = 0.40$) are fixed after calibration; sensitivity to alternative weight choices is not explored in this study.

In the full thesis, these findings will be extended with larger sample sizes, pair ablations, and broader task surfaces, but the poster already shows a stable message: **architecture design can buy back a large fraction of frontier-model quality without always paying frontier-model cost**.

SELECTED REFERENCES

- [1] Hao et al. **Hybrid SLM–LLM Inference**, 2024 · [2] Oh & Kim, **Uncertainty-Aware Hybrid Inference**, 2025 · [3] Shao et al. **Division-of-Thoughts**, 2025 · [4] Kwon et al. **vLLM: PagedAttention**, SOSP 2023 · [5] Hendrycks et al. **MMLU**, ICLR 2021 · [6] Clark et al. **ARC Challenge Set**, 2018 · [7] Cobbe et al. **GSM8K**, 2021 · [8] Zellers et al. **HellaSwag**, ACL 2019 · [9] Lin et al. **TruthfulQA**, ACL 2022 · [10] Companion SLR (54 studies, PRISMA 2020).

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